From Controversy to Consensus: Modelling Community-Sensitive Common Ground Management in German Discourse Markers

This study investigates three German discourse markers, which we label *consensus markers — soviel ich weiß* ('as far as I know'), *ja* ('roughly as you know'), and *bekanntlich* ('as is well known') that differ in the strength of consensus marking: ranging from weak epistemic constraint (*soviel ich weiß*) to strong public accessibility (*bekanntlich*). These markers are functionally similar: they mark a proposition p as uncontroversial [1], settled in the common ground [2], and can be used manipulatively to support downstream communicative goals such as convincing an interlocutor [3], often in contexts involving causal relationships [4]. However, previous work (cf. [5]) has mostly focused on *ja* and adopts a categorical framing that leaves open how notions like controversy vary *gradiently* across markers and interact with other contextual factors. Building on these insights, we propose a novel utility-based, noisy-threshold model within probabilistic pragmatics [6–8] and validate it with three preregistered experiments in social communication settings where causal information is relevant for convincing others to adopt a recommendation.[1]. Our theoretical contributions are threefold: (i) marker choice reflects a trade-off between perceived controversy and speaker goal strength, formalized as a utility function; (ii) perceived controversy further decomposes into propositional and pragmatic components, capturing the intuition that some controversial statements are less controversial within a specific community; (iii) the resulting utility space is partitioned by noisy thresholds rather than categorical boundaries, allowing overlapping regions where the decision is not clear-cut (see Fig. 2b). Empirically, we show that both perceived controversy and speaker goal strength have main effect on the production and comprehension of consensus markers (Fig. 2c–2d). Moreover, in production, the effect of goal strength is attenuated under high controversy. In comprehension, the strong marker *bekanntlich* may trigger pragmatic reactance [9]: listeners recognize the increased argumentative pressure and resist it, leading to lower adoption compared to the weaker *ja*. Overall, *ja* and *bekanntlich* function similarly — both are modulated primarily by perceived controversy, supporting a gradient view of boundaries between them. Our noisy-threshold model, trained on production data and applied to comprehension, captures these patterns within a single framework.

Model. Perceived controversy combines propositional and pragmatic components: $pc = pc_{\text{prop}} \cdot pc_{\text{prag}}$. Marker choice is governed by $U(pc, g) = -w_{pc} \cdot pc + w_g \cdot g - w_{int} \cdot pc \cdot g$, where g is goal strength and the interaction term captures the cost of invoking consensus under high controversy. Each marker u_i occupies a noisy interval $[\theta_i, \theta_{i+1})$ on the utility scale, with $P(u_i \mid pc, g) \propto \sigma(\lambda(U - \theta_i)) \cdot \sigma(-\lambda(U - \theta_{i+1}))$. A pragmatic listener inverts this model via Bayes' rule to infer pc and g, and downstream adoption depends on the inferred values.

Experiments and results. Three preregistered experiments validate the model (Fig. 1). In production (Fig. 2c), g increases marker strength (0.28, $P(> 0) = 1$), pc_{prag} decreases it (-0.17, $P(< 0) = 0.98$), and their interaction is negative (-0.22, $P(< 0) = 0.97$), as w_{int} predicts. In comprehension (Fig. 2d), both *ja* (0.49, $P(> 0) = 1$) and *bekanntlich* (0.46, $P(> 0) = 1$) increase inferred g, but only *ja* boosts adoption credibly (0.42, $P(> 0) = 0.99$; *bekanntlich*: 0.17, $P(> 0) = 0.90$). The marker-by-controversy interactions point in opposite directions for both inferences: for g, the *ja* effect trends larger under high controversy (0.28, $P(> 0) = 0.95$); for adoption, smaller (-0.31, $P(< 0) = 0.93$) — consistent with pragmatic reactance.

[1] Link to the pre-registration

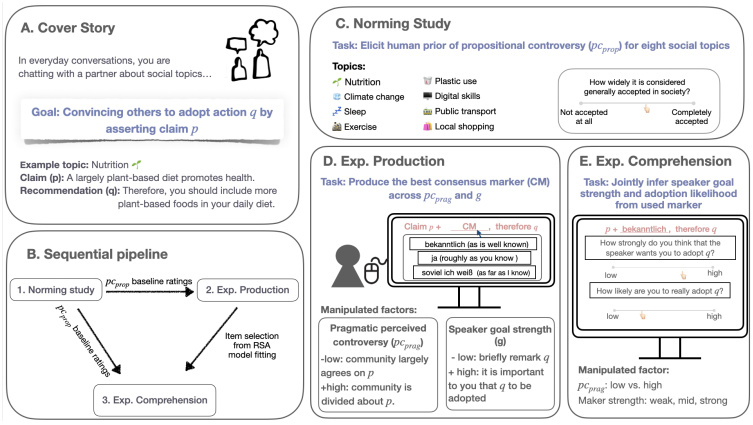
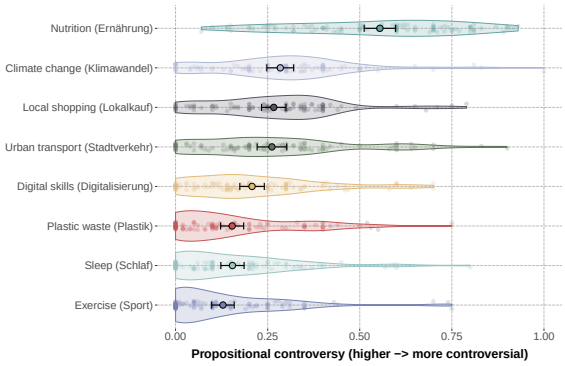
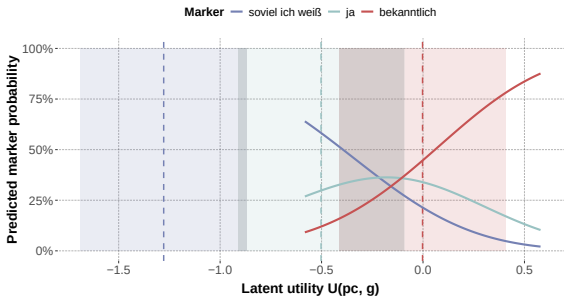


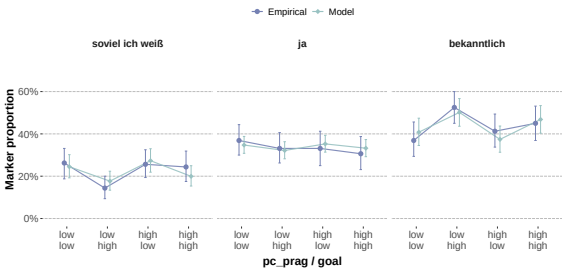
Figure 1: Experimental pipeline. (A) Cover story: participants discuss social topics with a partner, aiming to convince them to adopt a recommendation. (B) Sequential design: a norming study establishes baseline propositional controversy, followed by production and comprehension experiments. (C) Norming: eight topics rated for propositional controversy. (D) Production: participants select the best consensus marker given manipulated pragmatic controversy and goal strength. (E) Comprehension: participants infer speaker goal strength and adoption likelihood from marker-containing utterances.



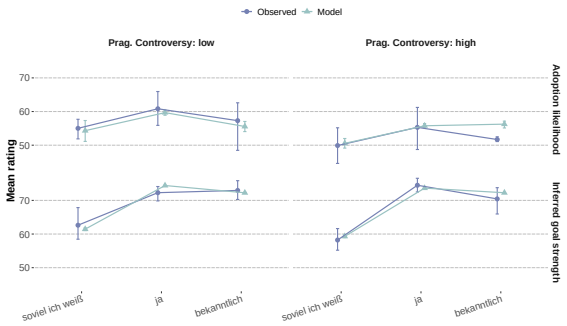
(a) Results of the norming experiment ($N = 102$).



(b) RSA noisy thresholds along the latent utility axis. Dashed lines: mean thresholds; shaded bands: overlapping regions.



(c) Results of the production experiment ($N = 80$).



(d) Results of the comprehension experiment ($N = 80$).

Figure 2: Model and experimental results. Points show means; error bars show bootstrap 95% confidence intervals.

References:

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